



SYMBIOSIS CENTRE FOR INFORMATION TECHNOLOGY (SCIT), PUNE

(A constituent of the Symbiosis International University, Pune)

T27 | IT HELPDESK TRIAGE BOT

AI-assisted ticket classification, prompt iteration and business impact analysis (Task T27 — Mini Project)

SEMESTER-I

AI Powered Tools

Submitted By:

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Build Evidence

The final application is a conversational IT helpdesk triage chatbot built in Google AI Studio. It uses an AI-safe ticket/playbook input, while the Gold Labels remain outside the AI system for Excel-based evaluation.

Build Link / Google AI Studio:

<https://aistudio.google.com/apps/b7b5377d-5199-4b32-897e-304d3b2ea6fc?showAssistant=true&showPreview=true>

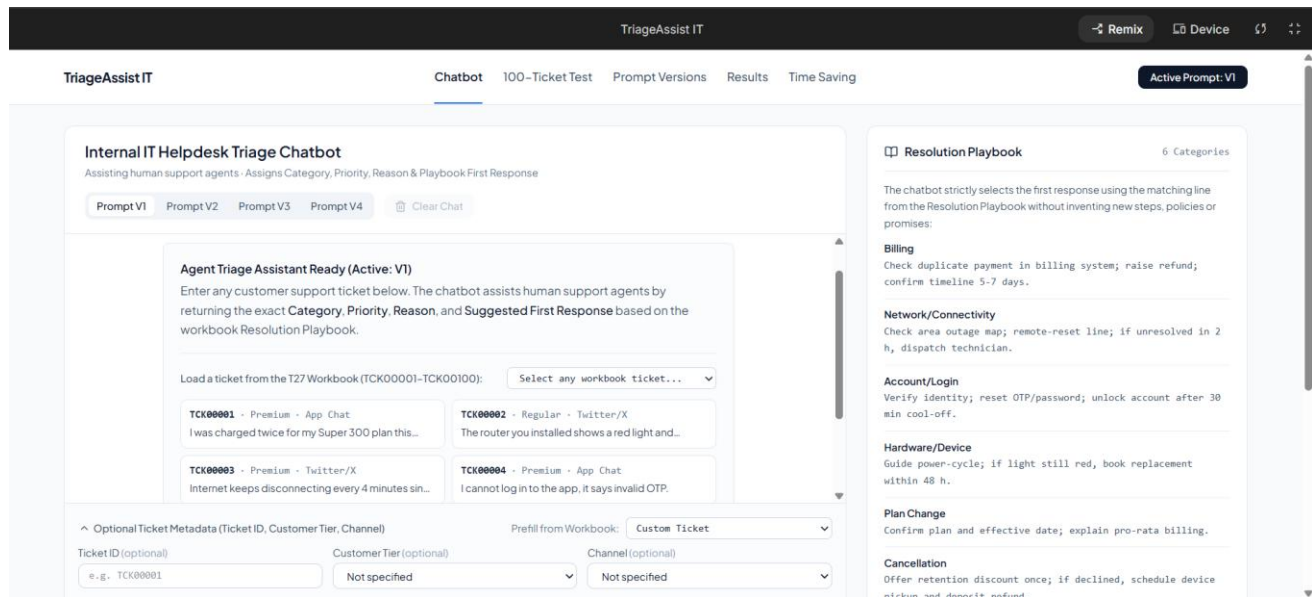


Figure 1 — Main chatbot interface with a completed ticket.

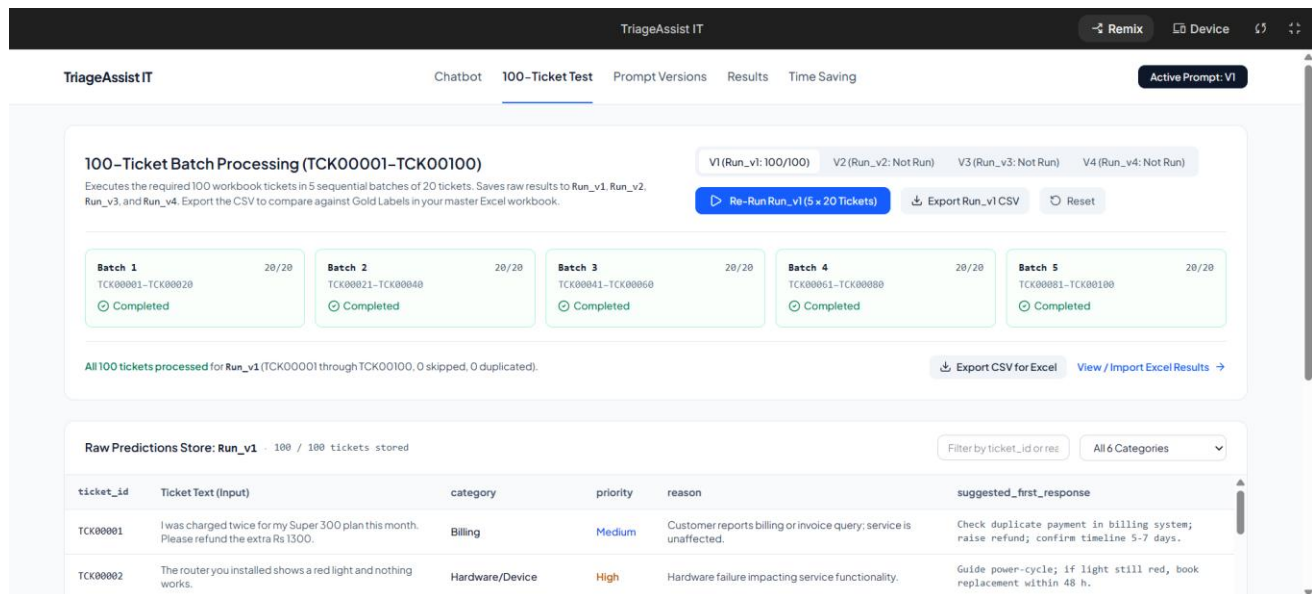


Figure 2 — 100-Ticket Test showing TCK00001-TCK00100 processed in five batches.

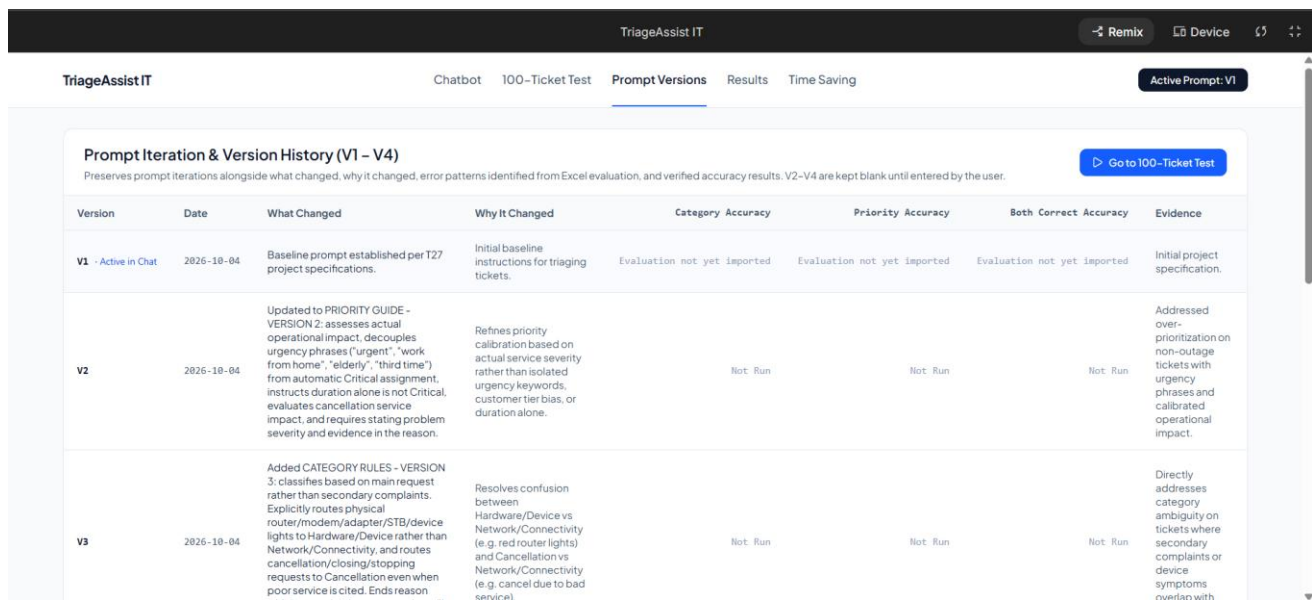


Figure 3 — Prompt Versions / Results showing V1-V4 and the selected final V3.

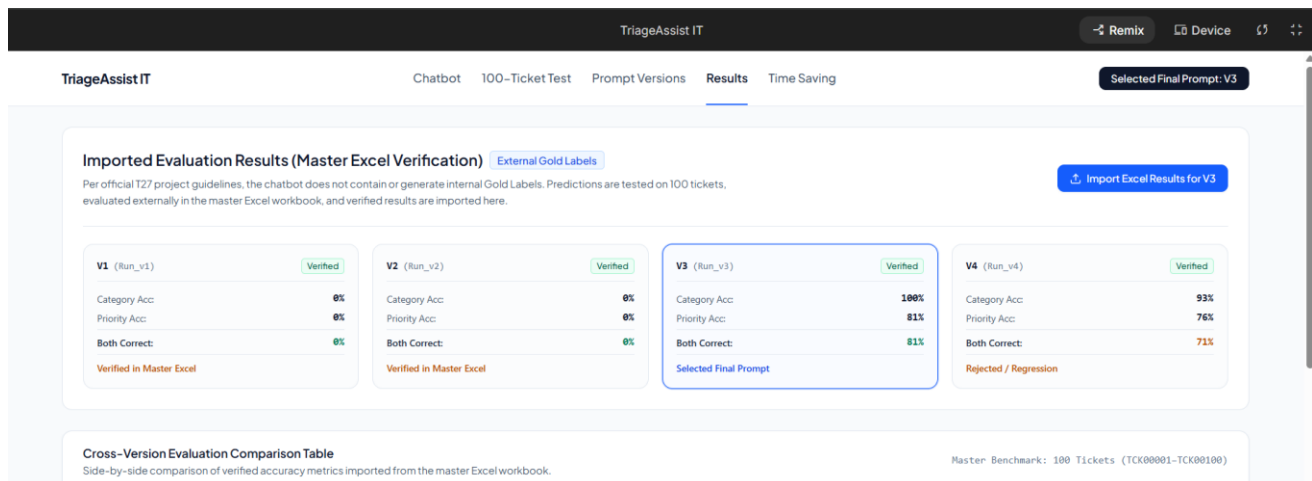


Figure 4 — Evaluation architecture and data separation: prediction first, Gold Label comparison afterward.

Methodology and Data Audit

Component	Method	Verification
AI input	Ticket ID, tier, channel, text + Resolution Playbook	Gold Labels excluded
Benchmark	TCK00001-TCK00100	Same 100 tickets in V1-V4
Prediction output	Ticket ID, category, priority, reason, playbook response	100 rows per run
Accuracy	Category, Priority, Both Correct	Excel vs Gold Labels
Final selection	Highest measured version	V3 retained; V4 regression documented

Table 1 — Methodology and data-audit summary

The master workbook contains the 500-ticket source data, Gold Labels, Resolution Playbook, AI Use Log and evaluation sheets. The chatbot uses a separate AI-safe input, preserving evaluation integrity.

15. Introduction

T27 – IT Helpdesk Triage Bot is a mini project under the Chatbots & AI Assistants theme. The objective is to read a support ticket, assign a category and priority, and suggest an approved first response from the Resolution Playbook. The same 100-ticket benchmark is then used for prompt iteration and verification.

15.62 Objectives

- Build a working conversational helpdesk triage chatbot.
- Evaluate V1-V4 on the same 100 tickets.
- Verify results systematically against the Gold Labels in Excel.
- Improve prompts using observed error patterns.
- Estimate agent time saved and make a manager-facing recommendation.

2. Dataset Overview and Data Quality

Item	Value
Size	500 support tickets × 7 columns
Benchmark	TCK00001-TCK00100 (100 tickets)
Channels	Twitter/X 132; Phone 130; Email 123; App Chat 115
Customer tiers	Regular 291; Premium 150; Business 59
Categories	Network/Connectivity 120; Billing 106; Hardware/Device 82; Account/Login 67; Cancellation 65; Plan Change 60
Priorities	High 268; Medium 149; Low 54; Critical 29
Playbook	6 approved category-specific responses
Data type	Synthetic teaching dataset

Table 2 — Dataset profile

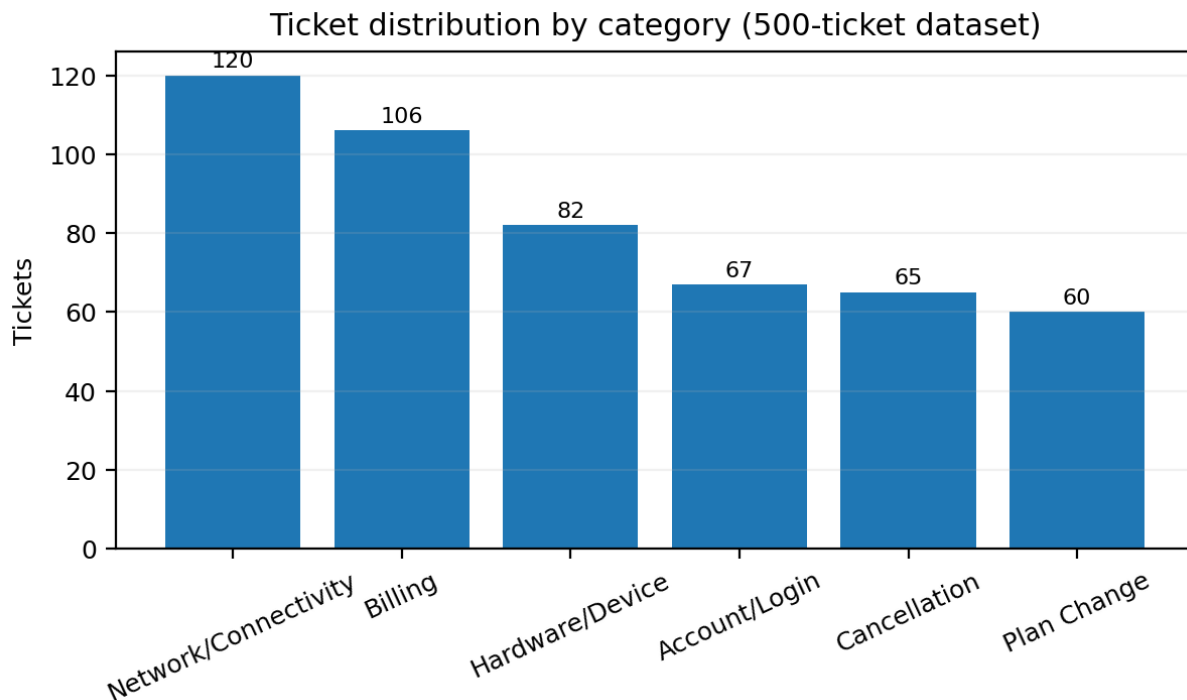


Figure 5 — Ticket distribution by category across the 500-ticket dataset.

Inference. Network/Connectivity and Billing are the largest categories, while Critical is comparatively rare. This makes class-specific evaluation important instead of relying only on overall accuracy.

3. AI-Assisted Development Process

Round	Problem found	Change / rationale	Outcome
V1	Starting point	Baseline category, priority, reason and playbook-response instructions.	93% cat.; 80% priority; 75% both.
V2	20 priority errors	Operational-impact and urgency calibration; urgency words alone do not force Critical.	100% cat.; 80% priority; 80% both.
V3	Category confusion observed	Two category boundary rules based on actual V1 errors; V2 priority guide retained.	100% cat.; 81% priority; 81% both.
V4	19 priority errors remained	Six worked examples + output validation.	93% cat.; 76% priority; 71% both. Rejected.

Table 3 — Documented prompt-improvement rounds (source: V1–V4 benchmark runs)

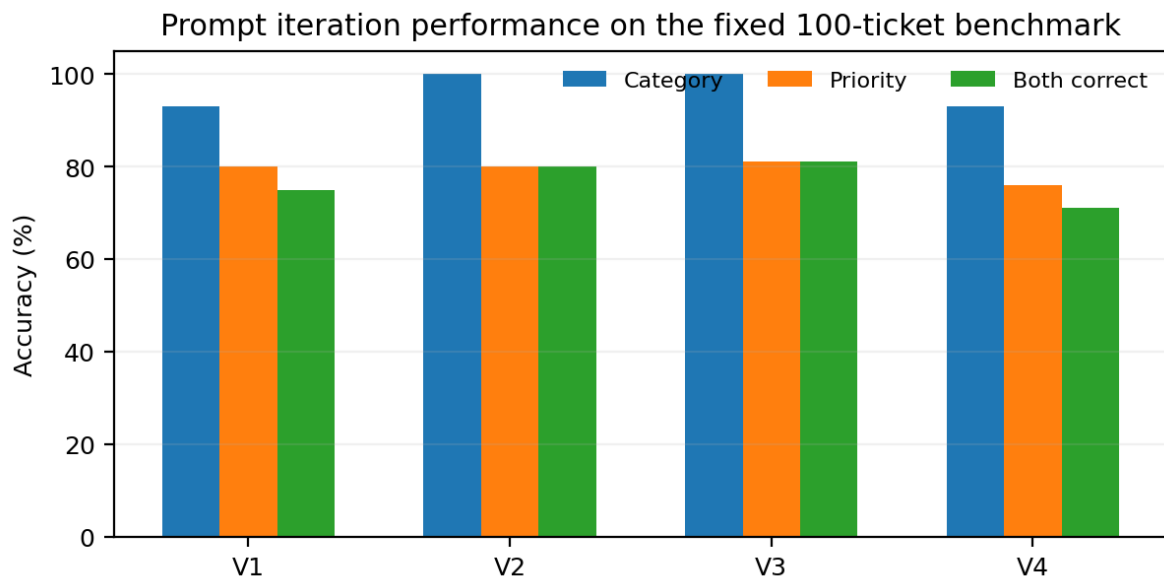


Figure 6 — Prompt iteration performance on the fixed benchmark.

The iteration was evidence-driven. V2 improved category classification; V3 retained that gain and slightly improved priority; V4 regressed and was rejected. The final selected version is V3.

4. Overall Analytical Results

Metric	V1	V2	V3	V4
Category Accuracy	93.0%	100.0%	100.0%	93.0%
Priority Accuracy	80.0%	80.0%	81.0%	76.0%
Both Correct Accuracy	75.0%	80.0%	81.0%	71.0%
Playbook compliance	100%	100%	100%	100%

Table 4 — Overall results by prompt version

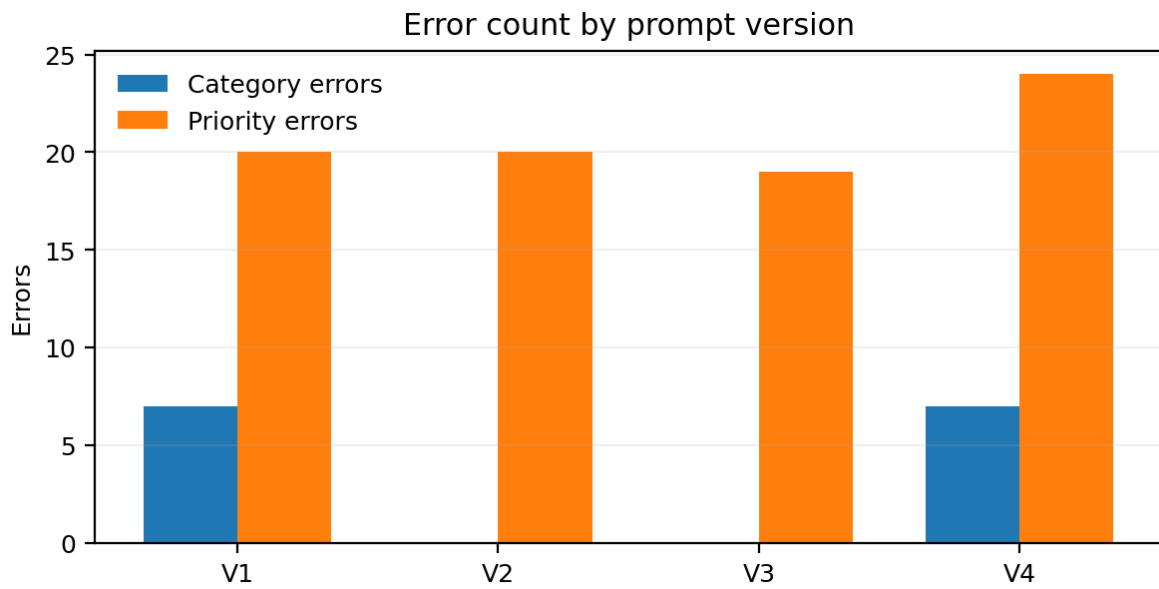


Figure 7 — Error count by prompt version.

5. V3 Final Performance

Priority	Correct	Total	Accuracy
Low	11	11	100.0%
Medium	25	25	100.0%
High	43	59	72.9%
Critical	2	5	40.0%

Table 5 — V3 priority accuracy by Gold Label priority

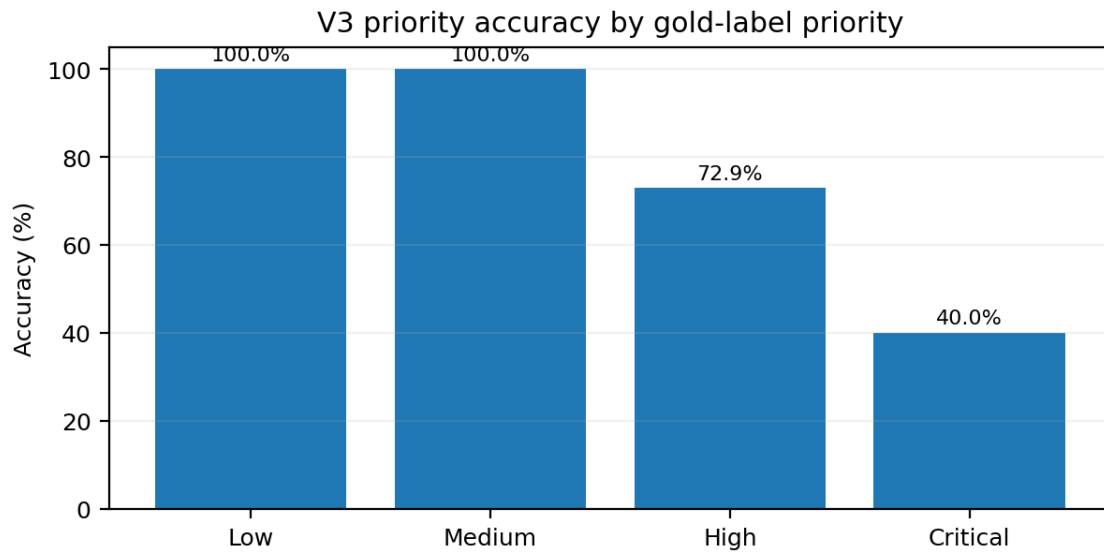


Figure 8 — V3 priority accuracy by Gold Label priority.

6. Priority Confusion and Business Interpretation

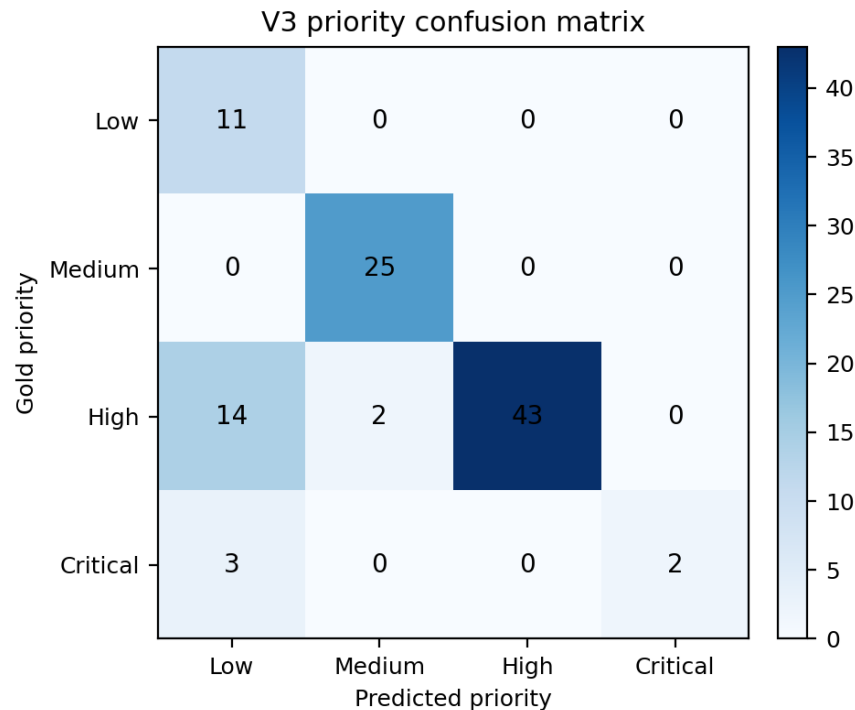


Figure 9 — V3 priority confusion matrix.

V3 correctly predicts all 11 Low and 25 Medium tickets. The main weakness is High-priority under-classification: 14 of 59 High tickets are predicted as Low and 2 as Medium. Three of 5 Critical tickets are predicted as Low.

7. Full-Dataset Business Analysis

Metric	Value
Tickets	500
Average resolution time	15.07 hours
Average CSAT	3.92 / 5
Resolution time vs CSAT	$r = -0.677$

Table 6 — Full-dataset business metrics

Segment	Tickets	Avg Resolution	Avg CSAT
Business	59	13.19 h	4.05
Premium	150	15.28 h	3.83
Regular	291	15.34 h	3.94
App Chat	115	15.40 h	3.84
Email	123	14.76 h	4.05
Phone	130	15.05 h	3.97
Twitter/X	132	15.08 h	3.81

Table 7 — Resolution time and CSAT by segment

Inference. Business customers have the lowest average resolution time (13.19 h) and highest CSAT (4.05). App Chat has the highest average resolution time (15.40 h), while Twitter/X has the lowest CSAT (3.81). The negative resolution-time/CSAT relationship suggests speed is an important service lever, but it is not causal proof.

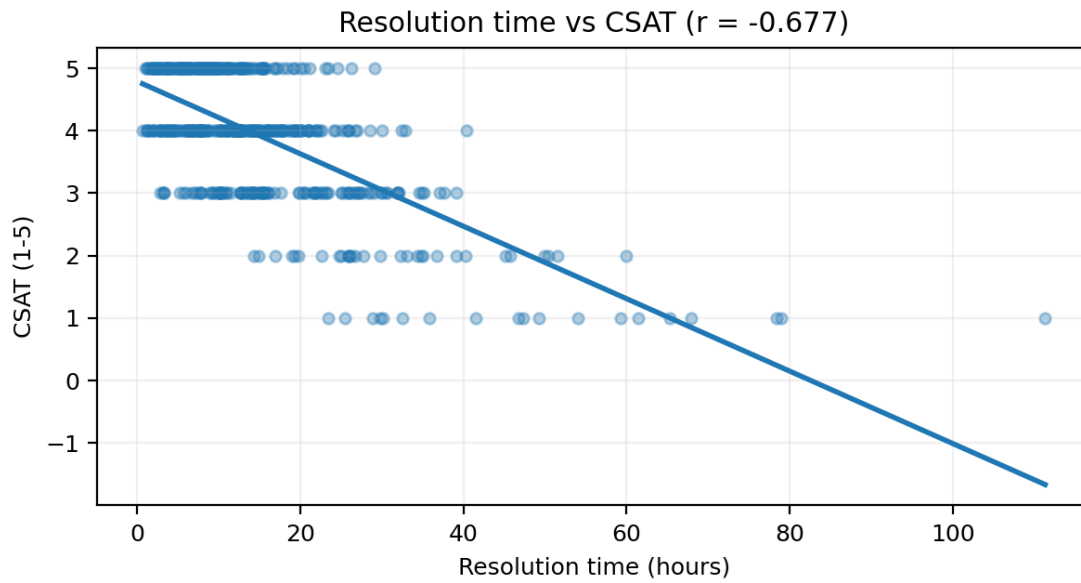


Figure 10 — Resolution time versus CSAT across the 500-ticket dataset.

8. Agent Time-Saving Estimate

Input	Value	Status
Manual triage time (A)	3.0 min/ticket	Assumed
Correct suggestion review time (B)	1.0 min/ticket	Assumed
Error correction time ©	2.0 min/ticket	Assumed
Final V3 both-correct accuracy	81.0%	Measured – Excel
Tickets per month	45.45	500 / 11 months
AI-assisted time	1.38 min/ticket	Calculated
Time saved	1.62 min/ticket	Calculated
Hours saved per month	1.23	Calculated
Hours saved per year	14.73	Calculated

Table 8 — Agent time-saving estimate — inputs and results

The estimate uses the project guide’s illustrative A/B/C assumptions. It is a scenario, not a measured productivity result.

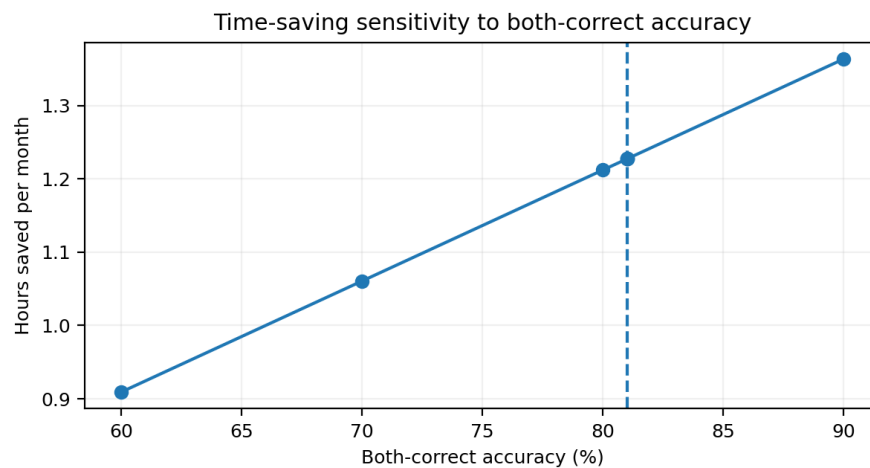


Figure 11 — Sensitivity of monthly savings to both-correct accuracy.

9. Business Recommendation

Action	Evidence	Metric
Use V3 as agent-assist, not autonomous triage	100% category; 81% both correct; priority remains weaker.	Both-correct and priority accuracy
Human review for priority, especially Critical	Critical accuracy 40.0%; High accuracy 72.9%.	Priority correction rate
Retain V3 after V4 regression	V4 fell to 71% both correct.	Accuracy after future prompt changes

Table 9 — Prioritised business recommendations

10. Verification and Correctness

Check	What was checked	Result
Benchmark completeness	TCK00001-TCK00100 processed in every run.	100/100 each run
Accuracy verification	Row-level comparison with Gold Labels in Excel.	Completed for V1-V4
Playbook compliance	Response matches playbook for predicted category.	100/100 each run
Prompt iteration	Changes tied to measured error patterns.	V3 selected
Time-saving arithmetic	A, B, C, T and V3 accuracy recomputed.	Reproduced

Table 10 — Verification evidence

11. Responsible AI and Limitations

- Synthetic teaching dataset; no real personal or confidential data was used.
- Gold Labels were not provided to the chatbot and were used only for post-run Excel verification.
- AI predictions were treated as suggestions and verified against the answer key.
- Priority accuracy is not sufficient for autonomous routing, especially for Critical tickets.
- Time-saving inputs are assumptions, not measured agent timings.
- V4 was retained as a documented failed iteration and rejected as the final model version.

12. Personal Reflection

The project showed that a working chatbot is not automatically a reliable chatbot. The V1 benchmark made the errors visible, and each later prompt was treated as an experiment rather than an assumption of improvement.

The strongest lesson was the value of verification. V3 achieved perfect category accuracy and the best combined performance, while V4 sounded more advanced but performed worse. The final decision was therefore based on evidence rather than on the complexity or recency of the prompt.

13. Contributions and AI Disclosure

Contributor / Tool	Contribution
Student	Prompt decisions, Excel verification, interpretation, recommendation and report review.
Google AI Studio / Gemini	Chatbot build and benchmark classification runs.
ChatGPT	Planning, prompt drafting, error analysis, business analysis and report structuring.
Excel	Gold Label verification, Accuracy, Error Log, Time Saving and Insights.

Table 11 — Contributors and AI disclosure

Generative AI use is disclosed in the project workbook. Reported benchmark results and business calculations were checked against the workbook before inclusion.